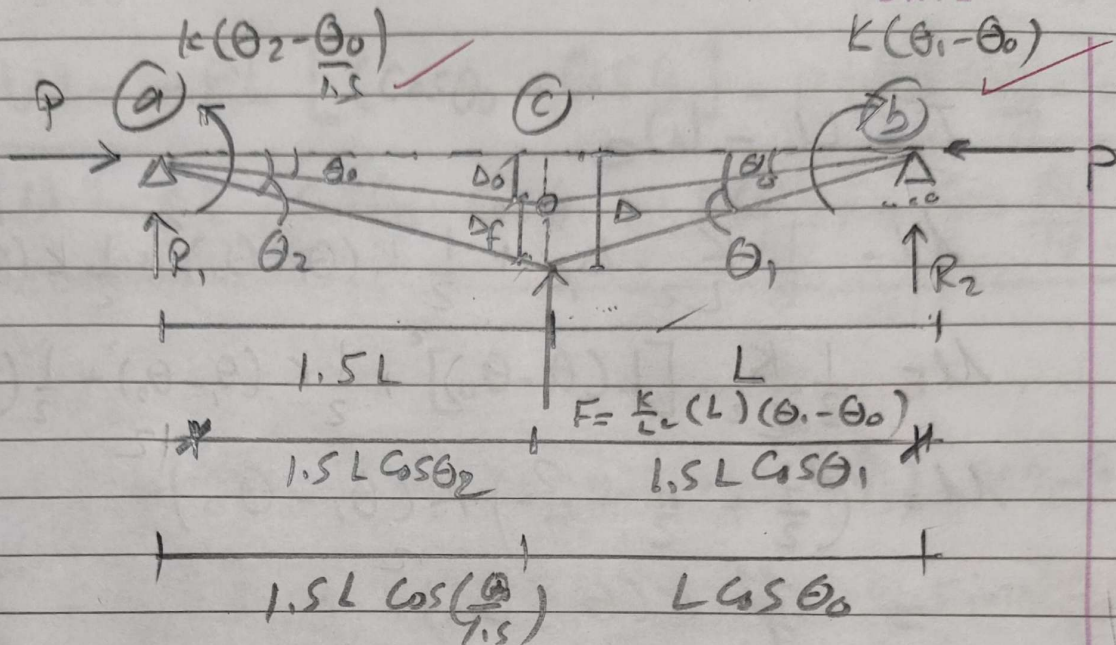


Pa

①

PAGE
DATE



① Energy approach

$$\sin \theta_1 = \frac{\Delta}{L}$$

$$\Delta = L \theta_1$$

$$\sin \theta_2 = \frac{\Delta}{1.5L}$$

$$\Delta = 1.5 \theta_2 L$$

$$\Delta = \Delta$$

$$L \theta_1 = 1.5 L \theta_2$$

$$\theta_1 = 1.5 \theta_2 \quad \text{--- (1)}$$

$$\Delta = \Delta_0 + \Delta_f$$

$$\sin \theta_0 = \frac{\Delta_0}{L}$$

$$\Delta_0 = L \theta_0 \quad \text{at point b}$$

$$\Delta_0 = L \frac{\theta_0}{1.5} \quad \text{at point a}$$

$$\Delta_f = \Delta - \Delta_0$$

$$\Delta_f = L \theta_1 - L \theta_0 = L (\theta_1 - \theta_0) \quad \text{--- (2)}$$

(2)

PAGE
DATE

$$\Pi = U - W_e$$

$$U = \frac{1}{2} \frac{K}{L^2} \cdot \Delta_f + \frac{1}{2} K (\theta_1 - \theta_0)^2 + \frac{1}{2} K \left(\theta_2 - \frac{\theta_0}{1.5} \right)^2$$

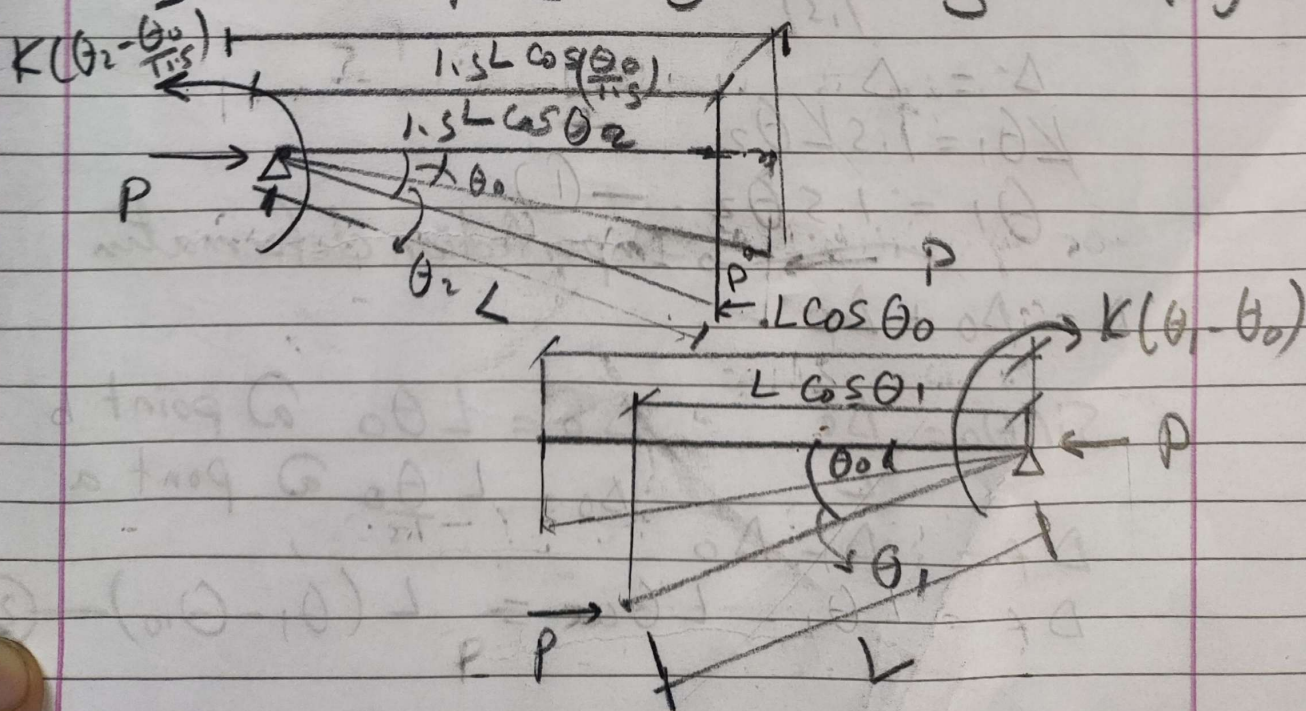
$$U = \frac{1}{2} \frac{K}{L^2} \cdot [L(\theta_1 - \theta_0)]^2 + \frac{1}{2} K (\theta_1 - \theta_0)^2 + \frac{1}{2} K \left(\frac{\theta_1}{1.5} - \frac{\theta_0}{1.5} \right)^2$$

$$U = \left(\frac{1}{2} + \frac{1}{2} + \frac{2}{9} \right) K (\theta_1 - \theta_0)^2$$

$$= \frac{11}{9} K (\theta_1 - \theta_0)^2$$

$$U = \frac{11}{9} K (\theta_1 - \theta_0)^2$$

$\Rightarrow W_e = -P$ (change in horizontal projection)



(3)

PAGE
DATE $\frac{2}{3} \theta_1$

$$W = -PL [\cos \theta_0 - \cos \theta_1] - PL \left[1.5 \cos \frac{\theta_0}{1.5} - \cos \frac{\theta_1}{1.5} \right]$$

$$W = -PL \left[\cos \theta_0 - \cos \theta_1 + 1.5 \cos \left(\frac{\theta_0}{1.5} \right) - \cos \frac{\theta_1}{1.5} \right]$$

$$\pi = U - W$$

$$= \frac{11}{9} k(\theta_1 - \theta_0)^2 - PL \left[\cos \theta_0 + 1.5 \cos \left(\frac{\theta_0}{1.5} \right) - \cos \theta_1 - \cos \left(\frac{\theta_1}{1.5} \right) \right]$$

$$\frac{d\pi}{d\theta_1} = \frac{22}{9} k(\theta_1 - \theta_0) - PL \left[\sin \theta_1 + \frac{2}{3} \sin \frac{\theta_1}{1.5} \right]$$

$$\sin \theta_1 = \theta_1$$

$$\sin \frac{\theta_1}{1.5} = \frac{\theta_1}{1.5}$$

$$\frac{d\pi}{d\theta_1} = \frac{22}{9} k(\theta_1 - \theta_0) - PL \left[\theta_1 + \frac{2}{3} \theta_1 \right] = 0$$

$$= \frac{22}{9} k(\theta_1 - \theta_0) - PL \left[\frac{5}{3} \theta_1 \right]$$

For equilibrium the total potential energy must assume stationary value this

$$\frac{d\pi}{d\theta} = 0$$

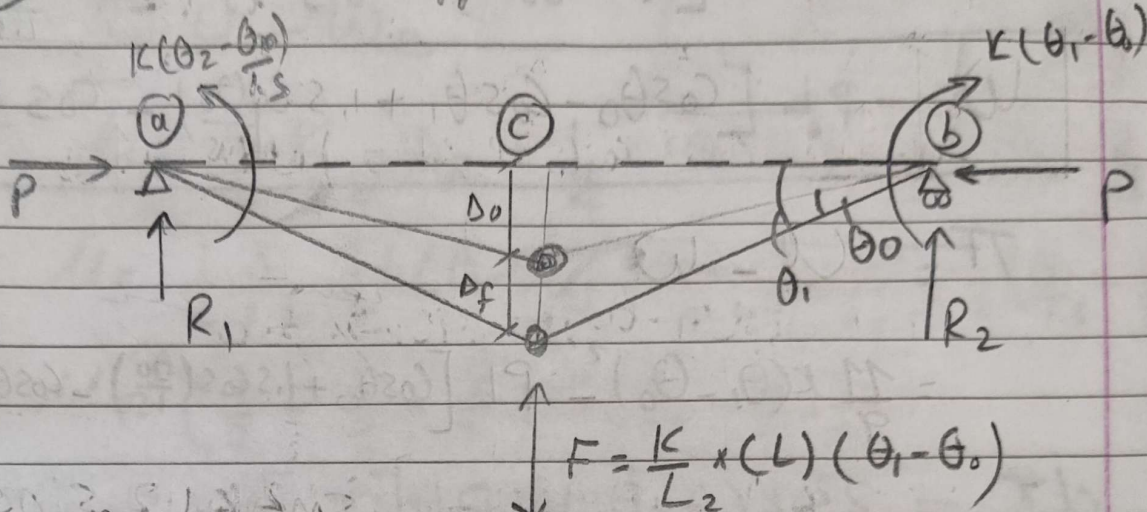
$$PL \left[\frac{5}{3} \theta_1 \right] = \frac{22}{9} k(\theta_1 - \theta_0)$$

$$P = \frac{22k}{15L} \left(1 - \frac{\theta_0}{\theta_1} \right)$$

(4)

PAGE
DATE

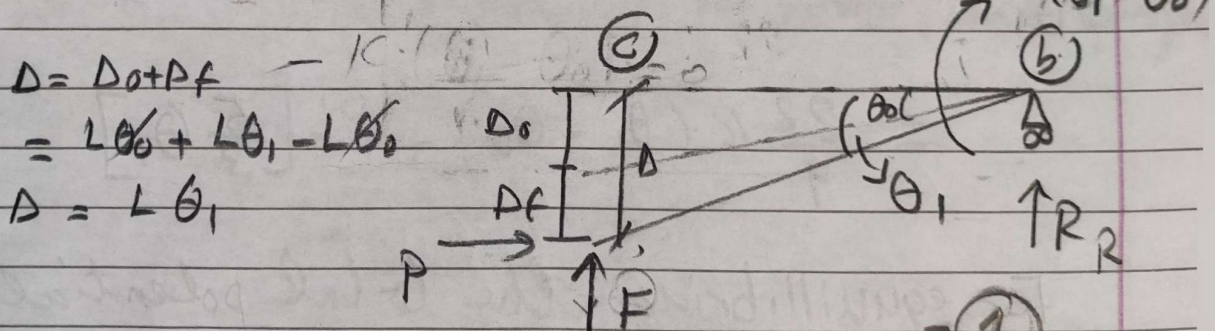
(2) Bifurcation Approach.



Find the value of Reaction R_1 & R_2

$$\sum M_c = 0$$

$$P D + R_2 L \cos \theta_1 - K (\theta_1 - \theta_0) = 0$$



$$\begin{aligned} D &= D_0 + P f \\ &= L \theta_0 + L \theta_1 - L \theta_0 \\ D &= L \theta_1 \end{aligned}$$

$$- P L \theta_1 + K (\theta_1 - \theta_0) = R_2 (L \cos \theta_1)$$

$$R_2 = \frac{K (\theta_1 - \theta_0) - P L \theta_1}{L}$$

(5)

PAGE
DATE $\sum M_0$ Not $\cos \theta_1 = 1$
 $\cos \theta_2 = 1$

$$\frac{k(\theta_1 - \theta_0) - PL\theta_1}{L} (1.5L \cos \theta_2 + L \cos \theta_1) + 1.5 \frac{k}{L} (\theta_1 - \theta_0) L \cos \theta_2$$

$$+ k \times \frac{2}{3} (\theta_1 - \theta_0) - k(\theta_1 - \theta_0) = 0$$

$$\frac{5}{2} k(\theta_1 - \theta_0) + \frac{3}{2} k(\theta_1 - \theta_0) + \frac{2}{3} k(\theta_1 - \theta_0) - \frac{3}{3} k(\theta_1 - \theta_0) - \frac{5}{2} PL\theta_1$$

$$\frac{11}{3} k(\theta_1 - \theta_0) = \frac{5}{2} PL\theta_1$$

$$\boxed{P_{cr} = \frac{22k}{15L} \left(1 - \frac{\theta_0}{\theta_1} \right)}$$

6

PAGE
DATE

⇒ Energy approach.

$$\sin \theta_1 = \frac{\Delta}{L} \Rightarrow \Delta = L \sin \theta_1$$

$$\sin \theta_2 = \frac{\Delta}{1.5L} \Rightarrow \Delta = 1.5L \sin \theta_2$$

$$\begin{aligned} 1.5L \sin \theta_2 &= L \sin \theta_1 \\ 1.5 \sin \theta_2 &= \sin \theta_1 \end{aligned}$$

also

$$L \sin \theta_1 - 1.5L \sin \theta_2 = 0$$

$$L \sin \theta_1 - 1.5L \sin \theta_2 = 0$$

$$\pi = U + V$$

Strain energy

$$\begin{aligned} U &= \frac{1}{2} k (\theta_1 - \theta_{01})^2 + \frac{1}{2} k (\theta_2 - \theta_{02})^2 \\ &\quad + \frac{1}{2} \left(\frac{k^2}{k_2} \right) [L \sin \theta_1 - \sin \theta_{01}]^2 \times k \end{aligned}$$

$$\begin{aligned} U &= \frac{1}{2} k (\theta_1 - \theta_{01})^2 + \frac{1}{2} k (\theta_2 - \theta_{02})^2 \\ &\quad + \frac{1}{2} k (\sin \theta_1 - \sin \theta_{01})^2 \end{aligned}$$

(7)

PAGE
DATEpotential of external load

$$V = -P[L(\cos\theta_{01} - \cos\theta_1) + 1.5L(\cos\theta_{02} - \cos\theta_2)]$$

$$\pi = \frac{1}{2}k(\theta_2 - \theta_{02})^2 + \frac{1}{2}k(\theta_1 - \theta_{01})^2 + \frac{1}{2}k(\sin\theta_1 - \sin\theta_{01})^2 - P[1.5L(\cos\theta_{02} - \cos\theta_2) + L(\cos\theta_{01} - \cos\theta_1)]$$

to differentiate π with respect to θ_1 :

First find a relationship between

$\frac{d\theta_2}{d\theta_1}$ to solve the above equation

because θ_2 is dependent in θ_1 , so

a small change in θ_1 causes change

in θ_2 .

$$\sin\theta_1 = 1.5 \sin\theta_2$$

$$\cos\theta_1 = 1.5 \cos\theta_2 \frac{d\theta_2}{d\theta_1}$$

$$\frac{d\theta_2}{d\theta_1} = \frac{\cos\theta_1}{1.5 \cos\theta_2}$$

(8)

PAGE
DATEfind $\frac{d\pi}{d\theta_1}$

$$\frac{d\pi}{d\theta_1} = K(\theta_2 - \theta_{02}) \frac{d\theta_2}{d\theta_1} + K(\theta_1 - \theta_{01})$$

$$+ K(\sin\theta_1 - \sin\theta_{01}) \cos\theta_1$$

$$- P \left[1.5L \sin\theta_2 \frac{d\theta_2}{d\theta_1} + L \sin\theta_1 \right] = 0$$

$$\Rightarrow K(\theta_2 - \theta_{02}) \left(\frac{\cos\theta_1}{1.5 \cos\theta_2} \right) + K(\theta_1 - \theta_{01}) + K(\sin\theta_1 - \sin\theta_{01}) \cos\theta_1$$

$$- P \left[1.5L \sin\theta_2 \left(\frac{\cos\theta_1}{1.5 \cos\theta_2} \right) + L \sin\theta_1 \right] = 0$$

$$P = \frac{K}{L} \times \frac{(\theta_2 - \theta_{02}) \left(\frac{\cos\theta_1}{1.5 \cos\theta_2} \right) + (\theta_1 - \theta_{01}) + (\sin\theta_1 - \sin\theta_{01}) \cos\theta_1}{\sin\theta_2 \left(\frac{\cos\theta_1}{\cos\theta_2} \right) + \sin\theta_1}$$

equilibrium equation

9

PAGE
DATE

post-buckling path

For the perfect system:-

$$\theta_{01} = 0 \quad \theta_{02} = 0$$

$$P = \frac{K}{L} \cdot \frac{\theta_2 \left(\frac{\cos \theta_1}{1.5 \cos \theta_2} \right) + \theta_1 + \sin \theta_1 \cos \theta_1}{\sin \theta_2 \left(\frac{\cos \theta_1}{\cos \theta_2} \right) + \sin \theta_1}$$

Check the stability

$\Delta \pi < 0$ unstable

$\Delta \pi > 0$ stable

In perfect equilibrium path

Choose a values for θ_0 , θ_1 , θ_2

$$\theta_{01} = 0.1 \text{ rad}$$

$$\sin \theta_{02} = \frac{\sin 0.1}{1.5}$$

$$\sin (0.10) = 0.0998334$$

$$\sin \theta_{02} = \frac{0.0998334}{1.5} = 0.0665556$$

(10)

PAGE
DATE

$$\theta_1 = 0.15 \text{ rad}$$

$$\sin \theta_2 = \frac{\sin 0.15}{1.5}$$

$$\sin 0.15 = 0.149438$$

$$\sin \theta_2 = \frac{0.149438}{1.5} = 0.0996253$$

$$\theta_2 = 0.09979 \text{ rad}$$

$$x = \frac{(\theta_1 - \theta_{01}) + (\sin \theta_1 - \sin \theta_{01}) \cos \theta_1 + \frac{(\theta_2 - \theta_{02}) \cos \theta_2}{1.5 \cos \theta_2}}{\sin \theta_1 + \sin \theta_2 \frac{\cos \theta_1}{\cos \theta_2}}$$

$$\theta_1 = 0.15 \quad \theta_{01} = 0.1 \quad \theta_2 = 0.09979 \quad \theta_{02} = 0.06666$$

Use Excel Sheet

$$x = 0.4872 \text{ @ } \theta_1 = 0.15$$

For first point on the path

(11)

PAGE
DATE

$$\pi = \frac{1}{2}(\theta_1 - \theta_{01})^2 + \frac{1}{2}(\theta_2 - \theta_{02})^2 + \frac{1}{2}(\sin \theta_1 - \sin \theta_{01})^2$$

$$- \times [\cos \theta_{01} - \cos \theta_1] + 1.5 [\cos \theta_{02} - \cos \theta_2]$$

$$\text{so } \theta_1 = 0.15$$

$$\pi(0.15) = +0.0000806$$

$$\pi > 0 \quad \text{stable.}$$

$\Rightarrow$ perfect post-buckling path.

$$\theta_{01} = 0$$

$$\theta_{02} = 0$$

$$P_{cr} = \frac{22}{15} = 1.4667$$

$$\theta_1 = 0 \quad \theta_2 = 0 \quad P = 1.4667$$

$$\pi(0) = 0$$

$$\theta_1 = 0.02$$

$$\pi(0.02) = -6.09 \times 10^{-9} < 0$$

unstable

Equilibrium paths computed from the derived equations

Given: $L = 5.0 \text{ m}$, $k = 200 \text{ kN}\cdot\text{m/rad}$, $\frac{k}{L} = 40 \text{ kN}$ $\bar{P} = \frac{PL}{k}$

Critical load: $P_{cr} = \frac{22k}{15L} = 58.67 \text{ kN}$ $\bar{P}_{cr} = \frac{P_{cr}L}{k} = \frac{22}{15} = 1.4667$

1 Compatibility (exact): $\sin \theta_1 = 1.5 \sin \theta_2 \Rightarrow \theta_2 = \arcsin\left(\frac{\sin \theta_1}{1.5}\right)$

2 Perfect post-buckling path (from derived equation):

$$\bar{P} = \frac{\theta_1 + \sin \theta_1 \cos \theta_1 + \frac{\theta_2 \cos \theta_1}{1.5 \cos \theta_2}}{\sin \theta_1 + \frac{\sin \theta_2 \cos \theta_1}{\cos \theta_2}}$$

This path is UNSTABLE

3 Imperfect equilibrium path (from derived equation):

$$\bar{P} = \frac{(\theta_1 - \theta_{10}) + (\sin \theta_1 - \sin \theta_{10}) \cos \theta_1 + \frac{(\theta_2 - \theta_{20}) \cos \theta_1}{1.5 \cos \theta_2}}{\sin \theta_1 + \frac{\sin \theta_2 \cos \theta_1}{\cos \theta_2}}$$

with $\sin \theta_{10} = 1.5 \sin \theta_{20}$

Numerical values used for the chart:

- Perfect path: $-0.45 \leq \theta_1 \leq 0.45 \text{ rad}$
- Imperfect paths: $\theta_{10} = +0.10 \text{ rad}$ and $\theta_{10} = +0.20 \text{ rad}$
- Negative side obtained by symmetry

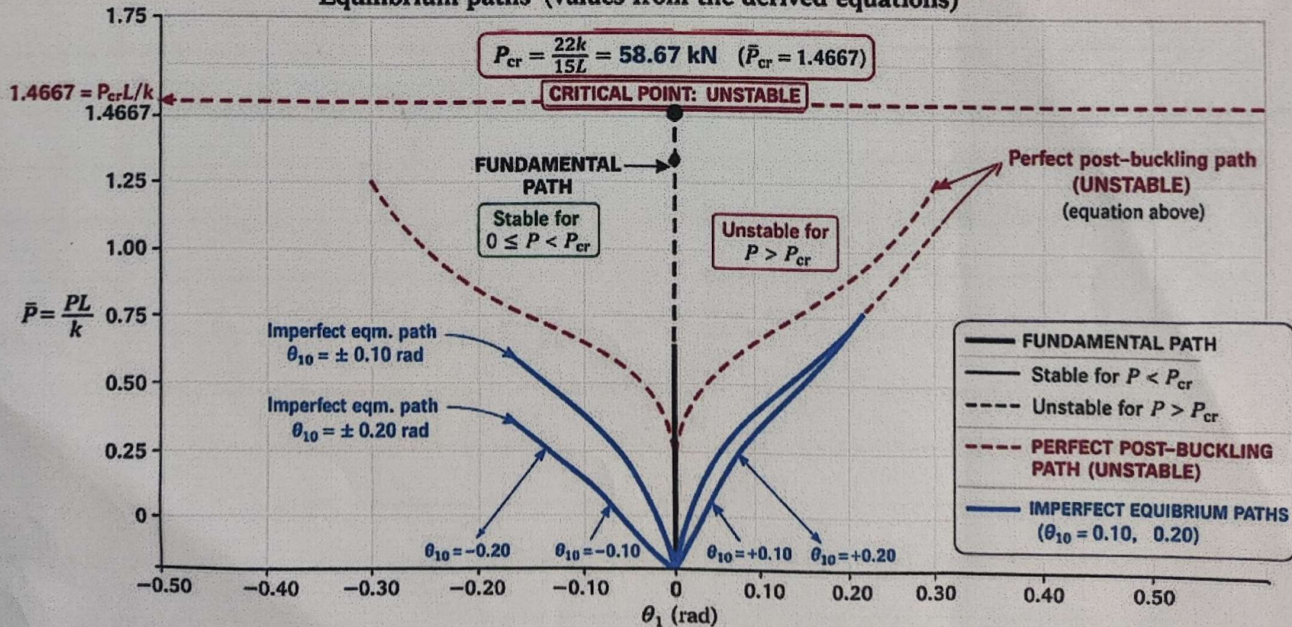
Perfect post-buckling path (UNSTABLE)

$\theta_1 \text{ (rad)}$	$\theta_2 \text{ (rad)}$	$\bar{P} = \frac{PL}{k}$	$P \text{ (kN)} = (k/L)\bar{P}$
0.00	0.0000	1.4667	58.67
0.10	0.0666	1.4658	58.63
0.20	0.1328	1.4631	58.52
0.30	0.1983	1.4588	58.35
0.40	0.2626	1.4531	58.12
0.45	0.2942	1.4499	58.00

Imperfect equilibrium path (STABLE response)

$\theta_1 \text{ (rad)}$	$\theta_2 \text{ (rad)}$	$\bar{P} \text{ for } \theta_{10} = 0.10$	$\bar{P} \text{ for } \theta_{10} = 0.20$	$P \text{ (kN)} = (k/L)\bar{P}$
0.10	0.0666	0.0000	—	0.00
0.15	0.0998	0.4872	—	19.49
0.20	0.1328	0.7297	0.0000	29.19
0.25	0.1657	0.8741	0.2900	34.96
0.30	0.1983	0.9692	0.4822	38.77
0.35	0.2306	1.0362	0.6183	41.45
0.40	0.2626	1.0853	0.7193	43.41
0.45	0.2942	1.1225	0.7968	44.90
0.50	0.3253	1.1514	0.8579	46.08

Equilibrium paths (values from the derived equations)



Dimensional load: $P = \bar{P} \frac{k}{L} = 40 \bar{P} \text{ kN}$ (since $k/L = 40 \text{ kN}$)

This plot is obtained strictly from the derived equations.
The critical point at $(\theta_1 = 0, P = P_{cr})$ is UNSTABLE, and the perfect post-buckling path is also UNSTABLE.

12

$$KD_1 + 2KD_2 - 3R_3 = 0$$

$$\Sigma F_y = 0$$

$$R_1 + R_2 - K\Delta_1 - K\Delta_2 = 0$$

$$R_1 + R_2 = K\Delta_1 + K\Delta_2$$

(13)

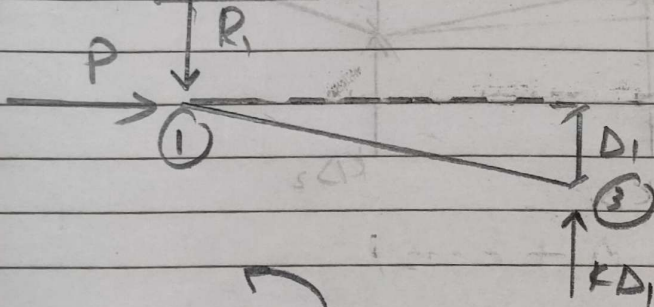
PAGE
DATE

$$R_1 + \frac{K}{3} (\Delta_1 + 2\Delta_2) = K\Delta_1 + K\Delta_2$$

$$R_1 = \frac{3K\Delta_1 + 3K\Delta_2 - K\Delta_1 - 2K\Delta_2}{3}$$

$$R_1 = \frac{2K\Delta_1 + K\Delta_2}{3}$$

$$R_1 = \frac{K}{3} (2\Delta_1 + \Delta_2)$$

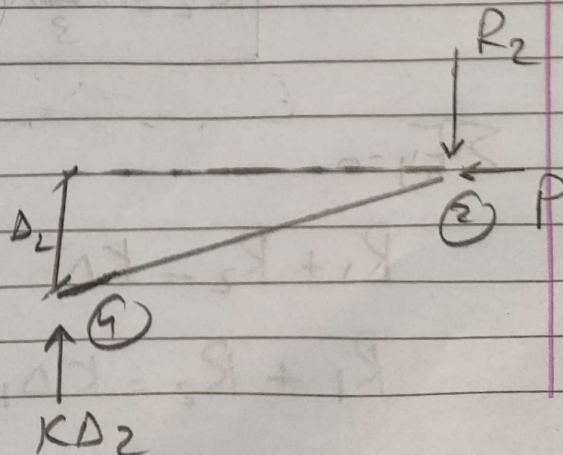

 $\sum M_2$

$$P\Delta_1 - R_1 L = 0$$

$$\Rightarrow R_1 = \frac{P\Delta_1}{L}$$

$$P\Delta_2 = R_2 L$$

$$\Rightarrow R_2 = \frac{P\Delta_2}{L}$$



(14)

PAGE
DATE

$$R_1 = \frac{K}{3} (2\Delta_1 + \Delta_2)$$

 $\downarrow$

$$\frac{P\Delta_1}{L} = \frac{K}{3} (2\Delta_1 + \Delta_2)$$

$$P\Delta_1 = \frac{2KL}{3} \Delta_1 + \frac{KL}{3} \Delta_2$$

$$\left[\left(P - \frac{2KL}{3} \right) \Delta_1 - \frac{KL}{3} \Delta_2 = 0 \right]$$

$$R_2 = \frac{K}{3} (\Delta_1 + 2\Delta_2)$$

$$\frac{P\Delta_2}{L} = \frac{K}{3} (\Delta_1 + 2\Delta_2)$$

$$P\Delta_2 = \frac{KL}{3} \Delta_1 + \frac{2KL}{3} \Delta_2$$

$$\left[-\frac{KL}{3} \Delta_1 + \left(P - \frac{2KL}{3} \right) \Delta_2 = 0 \right]$$

(15)

Solve the determinatePAGE
DATE

$$\begin{vmatrix} P - \frac{2KL}{3} & -\frac{KL}{3} \\ -\frac{KL}{3} & P - \frac{2KL}{3} \end{vmatrix} \begin{matrix} D_1 \\ D_2 \end{matrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}$$

$$\begin{vmatrix} P - \frac{2KL}{3} & -\frac{KL}{3} \\ -\frac{KL}{3} & P - \frac{2KL}{3} \end{vmatrix} = 0$$

$$\left(P - \frac{2KL}{3} \right)^2 - \left(\frac{KL}{3} \right)^2 = 0$$

$$\left(P - \frac{2KL}{3} - \frac{KL}{3} \right) \left(P - \frac{2KL}{3} + \frac{KL}{3} \right) = 0$$

$$(P - KL) \left(P - \frac{KL}{3} \right) = 0$$

$$P_1 = \frac{KL}{3} \quad P_2 = KL$$

$$\boxed{P_{cr} = \frac{KL}{3}}$$

(16)

PAGE
DATE

Buckling mode shape

$$\Rightarrow P = P_{cr} = \frac{KL}{3} \quad (\text{First mode})$$

$$\left(\frac{KL}{3} - \frac{2KL}{3}\right)\Delta_1 - \frac{KL}{3}\Delta_2 = 0$$

$$-\frac{KL}{3}\Delta_1 - \frac{KL}{3}\Delta_2 = 0$$

$$\Delta_1 + \Delta_2 = 0$$

$$\boxed{\Delta_2 = -\Delta_1} \quad \text{First mode}$$

$$\Rightarrow P = KL \quad (\text{Second mode})$$

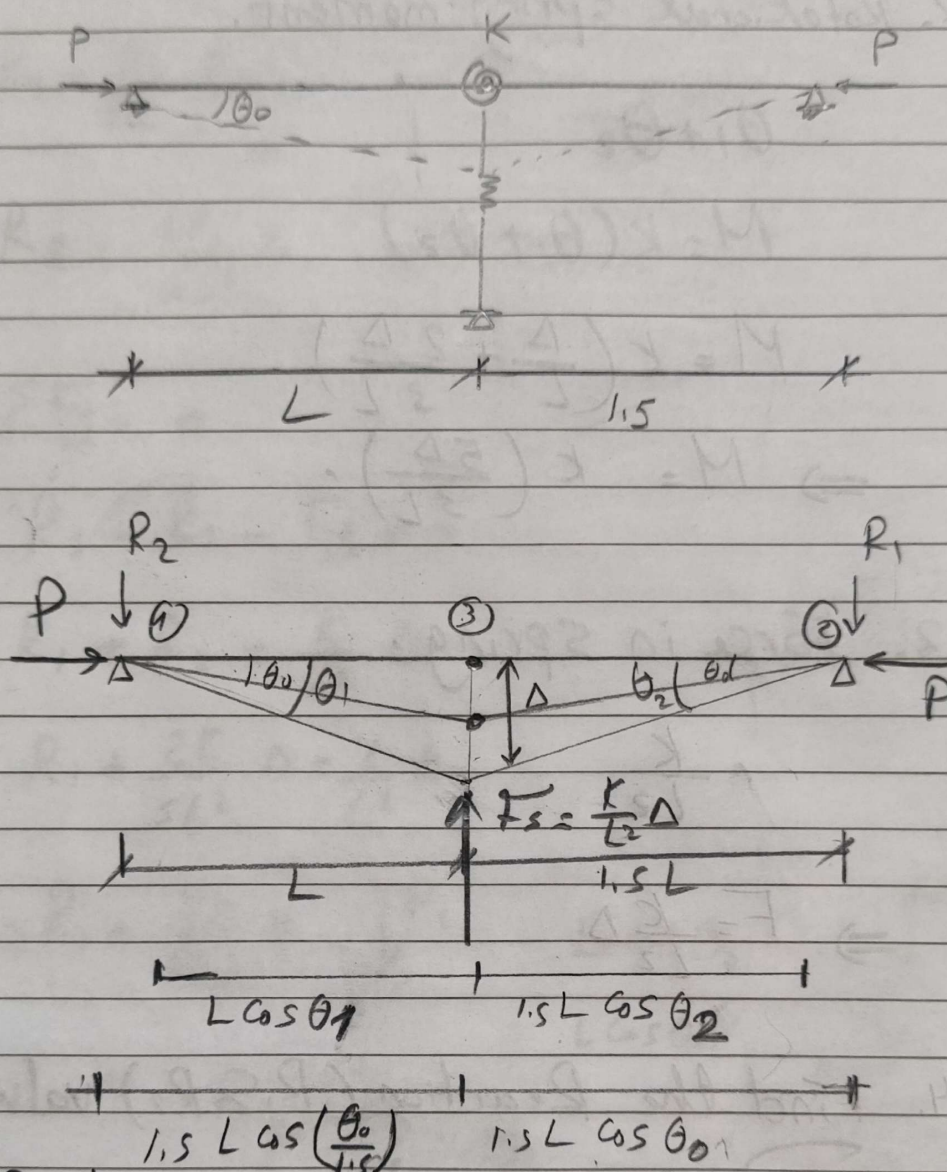
$$\left(KL - \frac{2KL}{3}\right)\Delta_1 - \frac{KL}{3}\Delta_2 = 0$$

$$\frac{KL}{3}\Delta_1 - \frac{KL}{3}\Delta_2 = 0$$

$$\boxed{\Delta_1 = \Delta_2}$$

PC

PAGE
DATE



① Relationship between θ_1, θ_2

$$\sin \theta_1 = \frac{\Delta}{L} \quad L \theta_1 = \Delta$$

$$\sin \theta_2 = \frac{\Delta}{1.5L} \quad 1.5L \theta_2 = \Delta$$

$$\Rightarrow 1.5 \theta_2 = \frac{2}{3} \theta_1$$

Solve
with imperfection

2. Rotational spring moment.

$$\theta_1 + \theta_2$$

$$M = K(\theta_1 + \theta_2)$$

$$M = K\left(\frac{\Delta}{L} + \frac{2\Delta}{3L}\right)$$

$$\Rightarrow M = K\left(\frac{5\Delta}{3L}\right)$$

3. Force in spring.

$$\frac{K}{L_2}$$

$$\Rightarrow F = \frac{K}{L_2} \Delta$$

4. Find the Reaction (R_1 & R_2) values

$$\sum M_i^+ = 0$$

$$F_s(L) - R_2\left(\frac{5L}{2}\right) = 0$$

$$\frac{RK}{L_2} \Delta \cdot L - R_2\left(\frac{5L}{2}\right) = 0$$

$$\frac{K}{L} \Delta - \frac{5L}{2} R_2 = 0$$

$$R_2 = \frac{2}{5L} \cdot \frac{k}{L} \Delta$$

$$\Rightarrow R_2 = \frac{2k}{5L^2} \Delta$$

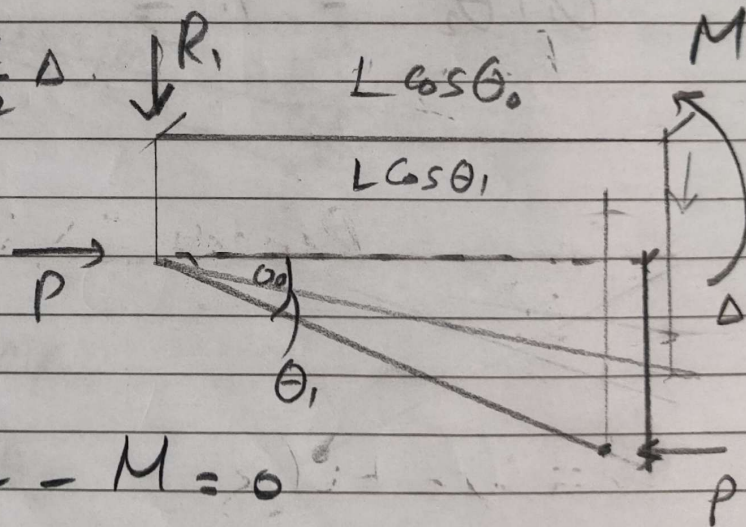
$$\sum F_y = 0$$

$$R_2 + R_2 - F_g = 0$$

$$R_1 + R_2 = \frac{k}{L^2} \Delta$$

$$R_1 + \frac{2k}{5L^2} \Delta = \frac{k}{L^2} \Delta$$

$$\Rightarrow R_1 = \frac{3k}{5L^2} \Delta$$



$$P\Delta - R_1 L - M = 0$$

$$P\Delta - \left(\frac{3k}{5L^2} \Delta\right) L - \frac{5k}{3L} \Delta = 0$$

$$\left[P - \frac{3k}{5L} - \frac{5k}{3L}\right] \Delta = 0$$

$$\left[P - \frac{34k}{15L} \right] \Delta = 0$$

$$P - \frac{34k}{15L} = 0$$

$$P_{cr} = \frac{34k}{15L}$$

5. Buckling mode

$$\theta_1 = \frac{\Delta}{L} \quad \theta_2 = \frac{2\Delta}{3L}$$

$$\theta_1 : \theta_2 = 1 : \frac{2}{3}$$

PC

PAGE
DATE

Solve the problem take into account imperfection

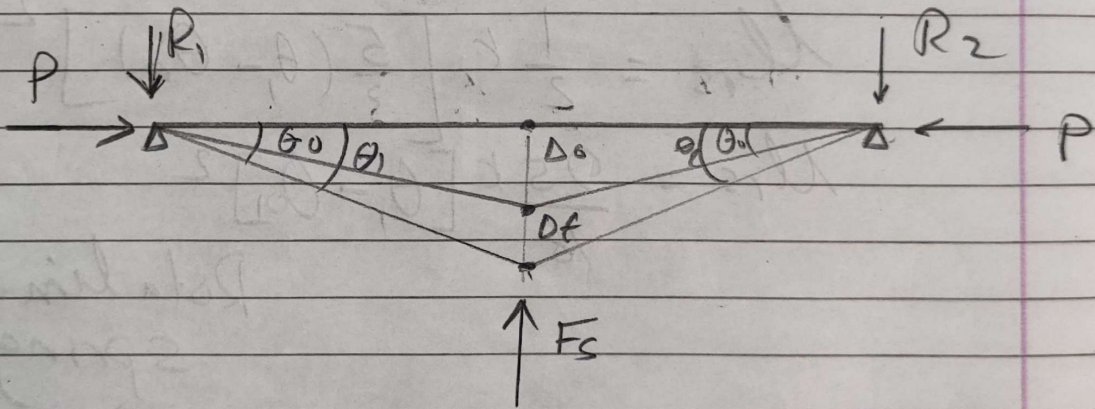
$$\sin \theta_1 = \frac{\Delta}{L} \quad \sin \theta_2 = \frac{\Delta}{1.5L}$$

for small deflection method

$$\sin \theta = \theta$$

$$\theta_1 = \frac{3}{2} \theta_2$$

$$\Rightarrow \theta_2 = \frac{2}{3} \theta_1$$



$$\Delta = \Delta_0 + \Delta_f$$

$$\frac{\Delta_0}{L} = \sin \theta_0$$

$$\Delta_0 = L \theta_0$$

$$\Delta_f = \Delta - \Delta_0 \Rightarrow L \theta_1 - L \theta_0 = L (\theta_1 - \theta_0)$$

Use energy approach

$$\Rightarrow U_s = \frac{1}{2} \left(\frac{k}{L^2} \right) [L(\theta_1 - \theta_0)]^2$$
$$= \frac{1}{2} k (\theta_1 - \theta_0)^2 \quad \text{Vertical spring energy}$$

$$\Rightarrow \theta_1 + \theta_2 = \theta_1 + \frac{2}{3}\theta_1 = \frac{5}{3}\theta_1$$

$$\theta_{01} + \theta_{02} = \theta_0 + \frac{2}{3}\theta_0 = \frac{5}{3}\theta_0$$

$$U_{rs} = \frac{1}{2} k \left[\frac{5}{3} (\theta_1 - \theta_0) \right]^2$$

$$U_{rs} = \frac{25k}{18} [\theta_1 - \theta_0]^2$$

Rotational
spring
energy

$$U = U_s + U_{rs}$$

$$= \frac{1}{2} k (\theta_1 - \theta_0)^2 + \frac{25}{18} k (\theta_1 - \theta_0)^2$$

$$U = \frac{17k}{9} (\theta_1 - \theta_0)^2$$

$$W = -PL [\cos \theta_0 - \cos \theta_1] - 1.5 PL \left[\cos \left(\frac{2}{3} \theta_1 \right) - \cos \left(\frac{2}{3} \theta_0 \right) \right]$$

$$\pi = \frac{17}{9} k (\theta_1 - \theta_0)^2 - PL \left[\cos \theta_1 + 1.5 \cos \left(\frac{2}{3} \theta_1 \right) - \cos \theta_0 - 1.5 \cos \left(\frac{2}{3} \theta_0 \right) \right]$$

$$\frac{d\pi}{d\theta_1} =$$

$$\frac{d\pi}{d\theta_1} = \frac{34}{9} k (\theta_1 - \theta_0) - PL \left[\sin \theta_1 + 1.5 \sin \left(\frac{2}{3} \theta_1 \right) \right] \times \frac{1}{1.5}$$

$$= \frac{34}{9} k (\theta_1 - \theta_0) - PL \left[\theta_1 + \frac{2}{3} \theta_1 \right]$$

$$= \frac{34}{9} k (\theta_1 - \theta_0) - \frac{5}{3} PL \theta_1 = 0$$

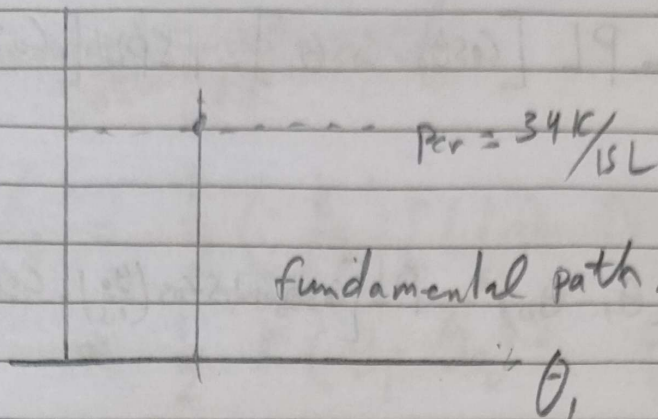
$$P = \frac{34 k}{15L} \frac{\theta_1 - \theta_0}{\theta_1}$$

* fundamental path and post-buckling path in small deflection -

$$\theta_1 = 0 \quad \theta_2 = 0$$

$$P = \frac{34 k}{15L}$$

P



fundamental path.

Check the stability for fundamental path perfect column -

$$\Pi = \frac{17k}{9} (\theta_1 - \theta_{01})^2 - \frac{5PL}{6} (\theta_1^2 - \theta_{01}^2)$$

$\theta_{01} = 0 \Rightarrow$ perfect column

$$\Pi = \left(\frac{17k}{9} - \frac{5PL}{6} \right) \theta_1^2$$

$$\frac{d\Pi}{d\theta} = 2 \left(\frac{17k}{9} - \frac{5PL}{6} \right) \theta_1$$

$$\frac{d^2\Pi}{d\theta^2} = 2 \left(\frac{17k}{9} - \frac{5PL}{6} \right)$$

stable fundamental path -

$$\frac{17k}{9} - \frac{5PL}{6} > 0$$

$$P < \frac{34k}{18L}$$

Unstable fundamental path

$$\frac{17k}{9} - \frac{5PL}{6} < 0$$

$$P > \frac{34k}{15L}$$

Check the stability for fundamental path in perfect equilibrium.

$$\pi = \frac{17k}{9}(\theta_2 - \theta_{01})^2 - \frac{5PL}{6}(\theta_1^2 - \theta_{01}^2)$$

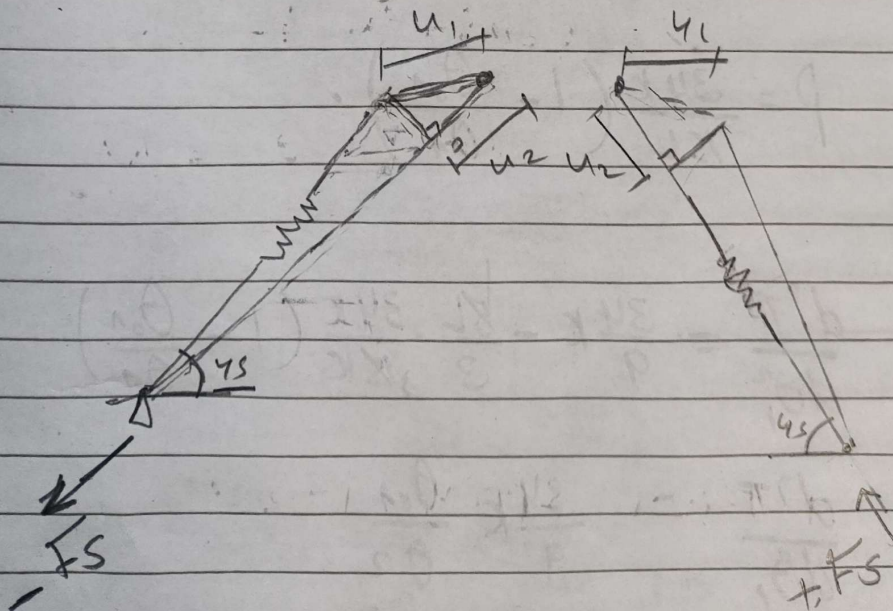
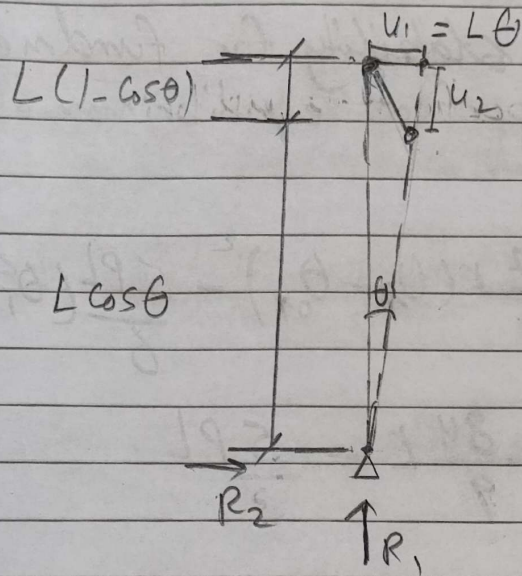
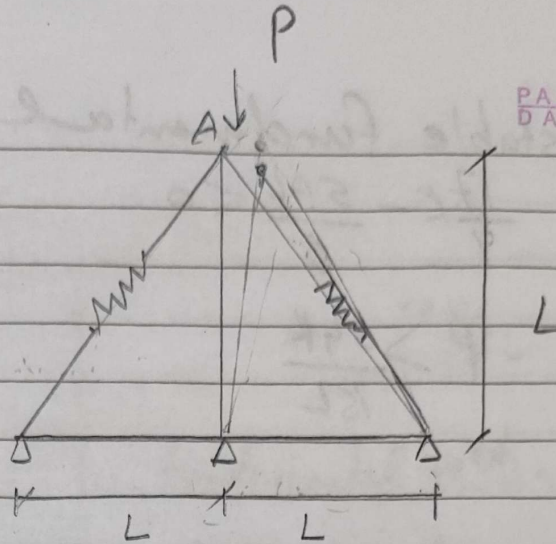
$$\frac{d^2\pi}{d\theta_1^2} = \frac{34k}{9} - \frac{5PL}{3}$$

$$P = \frac{34k}{15L} \left(1 - \frac{\theta_{01}}{\theta_1}\right)$$

$$\frac{d^2\pi}{d\theta_1^2} = \frac{34k}{9} - \frac{5L}{3} \cdot \frac{34k}{15L} \left(1 - \frac{\theta_{01}}{\theta_1}\right)$$

$$\frac{d^2\pi}{d\theta_1^2} = \frac{34k}{9} \frac{\theta_{01}}{\theta_1^2}$$

$$\frac{d^2\pi}{d\theta_1^2} > 0$$



* Coordinates system

* the top joint A moves.

* horizontally

$$u_1 = L \sin \theta$$

* vertically downward by

$$u_2 = L(1 - \cos \theta)$$

$$\Rightarrow u = (u_1, -u_2)$$

$\Rightarrow$ left spring elongation

initial coordinate for each direction

$$\Delta_1 = u_1 \sin 45 - u_2 \cos 45$$

$$\Delta_1 = \frac{u_1 - u_2}{\sqrt{2}}$$

$\Rightarrow$ Right spring elongation

$$\Delta_2 = \frac{-u_1 - u_2}{\sqrt{2}}$$

Strain energy

$$U^2 = \frac{1}{2} K \Delta^2$$

$$U = \frac{1}{2} K \Delta_1^2 + \frac{1}{2} K \Delta_2^2$$

$$U = \frac{1}{2} K \left(\frac{u_1 - u_2}{\sqrt{2}} \right)^2 + \frac{1}{2} K \left(\frac{-u_1 - u_2}{\sqrt{2}} \right)^2$$

$$U = \frac{K}{4} [(u_1 - u_2)^2 + (u_1 + u_2)^2]$$

$$(a-b)^2 + (a+b)^2 = 2a^2 + 2b^2$$

$$U = \frac{K}{2} (2u_1^2 + 2u_2^2)$$

$$U = \frac{K}{2} (u_1^2 + u_2^2)$$

$$u_1 = L \sin \theta \quad u_2 = L(1 - \cos \theta)$$

$$U = \frac{KL^2}{2} [\sin^2 \theta + (1 - \cos \theta)^2]$$

$$W = -PL(1 - \cos \theta)$$

$$\Pi = \frac{KL^2}{2} [\sin^2 \theta + (1 - \cos \theta)^2] - PL(1 - \cos \theta)$$

$$\frac{d\Pi}{d\theta} = 0$$

$$\frac{KL^2}{2} [2 \sin \theta \cos \theta + 2(1 - \cos \theta) \cdot \sin \theta]$$

$$KL^2 \sin \theta = PL \sin \theta = 0$$

$$KL^2 - PL = 0$$

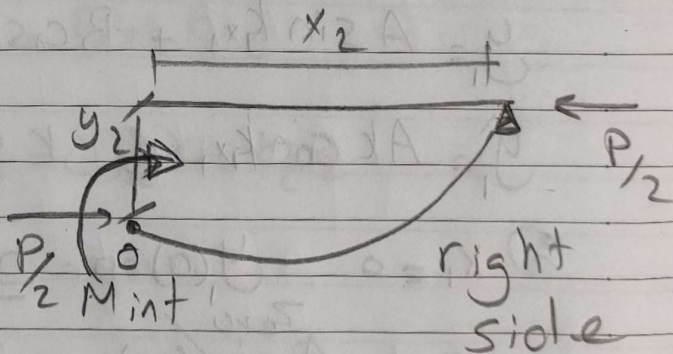
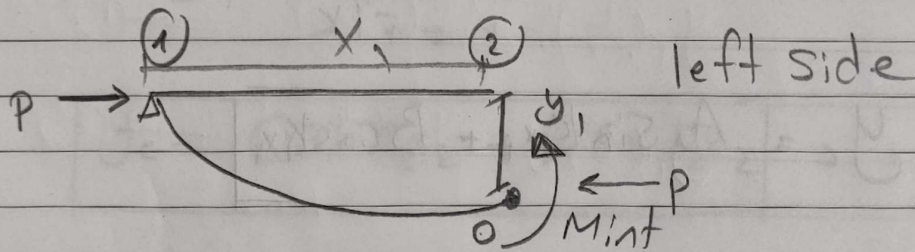
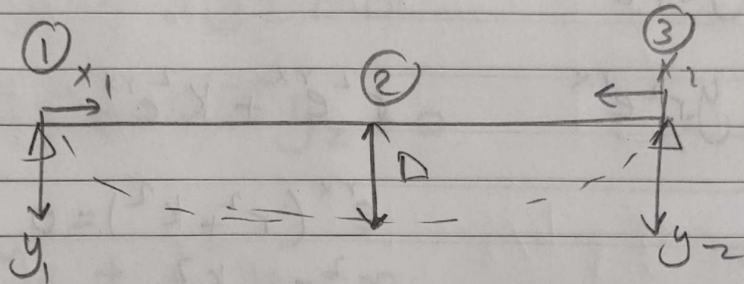
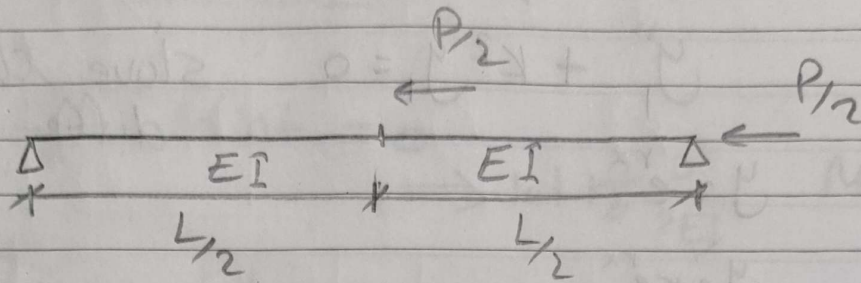
$$\Rightarrow \boxed{P = KL}$$

①

التاريخ:

الموضوع:

Q1



For the left side

$$\sum M_{a0} = 0$$

$$P y_1 - M_{int} = 0$$

$$\Rightarrow \text{let } K_1^2 = \frac{P}{EI}$$

$$M_{int} = -EI y''$$

(2)

التاريخ:

الموضوع:

$$EI y_1'' + K^2 EI y_1 = 0$$

$$y_1'' + K^2 y_1 = 0$$

solve the
differ equation

$$y = e^{rx}$$

$$y' = r e^{rx}$$

$$y = r^2 e^{rx}$$

$$r^2 e^{rx} + K^2 e^{rx} = 0$$

$$e^{rx} (r^2 + K^2) = 0$$

$$r^2 = -K^2$$

$$r = \pm iK$$

$$y_c = A \sin Kx_1 + B \cos Kx_1$$

$$y_1 = A \sin Kx_1 + B \cos Kx_1$$

$$y_1' = Ak \cos Kx_1 - BK \sin Kx_1$$

at $x_1 = 0$ $y(0) = 0$ no deflection

$$0 = A \overset{\text{Zero}}{\cancel{\sin(0)}} + B \overset{1}{\cancel{\cos(0)}}$$

$$B = 0$$

the left side

(3)

التاريخ:

الموضوع:

For the right side

$$\frac{P}{2} y_2 - M_{int} = 0 \Rightarrow \text{let } k^2 = \frac{P}{2EI} \quad M_{int} = -EI y_2''$$

$$2EI y_2'' + P y_2 = 0$$

$$2EI y_2'' + k^2 EI y_2 = 0$$

$$\Rightarrow y_2'' + k^2 y_2 = 0$$

$$y_c = C \sin k x_2 + D \cos k x_2$$

$$y_2 = C \sin k x_2 + D \cos k x_2$$

$$y_2' = C k \sin k x_2 - D k \cos k x_2$$

$$\text{at } x_2 = 0 \quad y_2(0) = 0$$

$$D = 0$$

to find the other constant A & C

Use compatibility condition at the middle.

4

التاريخ:

الموضوع:

from $k_1^2 = \frac{P}{EI}$ and $k_2^2 = \frac{P}{2EI}$

$$k_2^2 = \frac{k_1^2}{2} \Rightarrow \boxed{k_2 = \frac{k_1}{\sqrt{2}}}$$

Remaining 2 Constant we need
2 B/c

a) $x_1 = x_2 = L/2$

$$y_1\left(\frac{L}{2}\right) = y_2\left(\frac{L}{2}\right) \quad \text{--- (1)}$$

$$y_1\left(\frac{L}{2}\right) = -y_2\left(\frac{L}{2}\right) \quad \text{--- (2)}$$

equation No1

$$A \sin \frac{k_1 L}{2} = C \sin \frac{k_2 L}{2}$$

$$A \sin \frac{k_1 L}{2} - C \sin \frac{k_2 L}{2} = 0$$

equation No2

$$A k_1 \cos \frac{k_1 L}{2} = -C k_2 \cos \frac{k_2 L}{2}$$

$$A k_1 \cos \frac{k_1 L}{2} + C k_2 \cos \frac{k_2 L}{2} = 0$$

(5)

القاريخ:

الموضوع:

From the pre equation write it in matrix form to have the solution

$$\begin{vmatrix} \sin \frac{k_1 L}{2} & -\sin \frac{k_2 L}{2} \\ k_1 \cos \frac{k_1 L}{2} & k_2 \cos \frac{k_2 L}{2} \end{vmatrix} \begin{vmatrix} A \\ C \end{vmatrix} = \begin{vmatrix} 0 \\ 0 \end{vmatrix}$$

$$k_2 \sin \frac{k_1 L}{2} \cos \frac{k_2 L}{2} + k_1 \cos \frac{k_1 L}{2} \sin \frac{k_2 L}{2} = 0$$

$$k_1 = \sqrt{2} k_2$$

$$\Rightarrow -\cancel{k_1} \sin \frac{k_2 L}{\sqrt{2}} \cos \frac{k_2 L}{2} + \sqrt{2} k_2 \cos \frac{k_2 L}{\sqrt{2}} \sin \frac{k_2 L}{2} = 0$$

divide everything by

$$\cos\left(\frac{k_2 L}{\sqrt{2}}\right) \cos\left(\frac{k_2 L}{2}\right)$$

$$\Rightarrow \left[\tan\left(\frac{k_2 L}{\sqrt{2}}\right) + \sqrt{2} \tan\left(\frac{k_2 L}{2}\right) = 0 \right]$$

(6)

التاريخ:

الموضوع:

x (Kil)	$f(x)$
0.1	0.141
0.6	0.3741
1.1	1.1248
1.6	1.0621
2.1	0.6009
2.6	-0.102
3.1	-0.806

— solve the equation

$$\Rightarrow X = 2.531$$

$$K_L = 2.531 = \sqrt{\frac{P}{12EI}} \cdot L$$

$$(6.405961)(2) EI = P \cdot L^2$$

$$\Rightarrow \boxed{P_{cr} = \frac{12.812 EI}{L^2}}$$

$$K_e = \sqrt{\frac{P_e}{P_{cr}}} = \sqrt{\frac{\pi^2 EI / L^2}{12.8 EI / L^2}}$$

$$= \boxed{0.88}$$

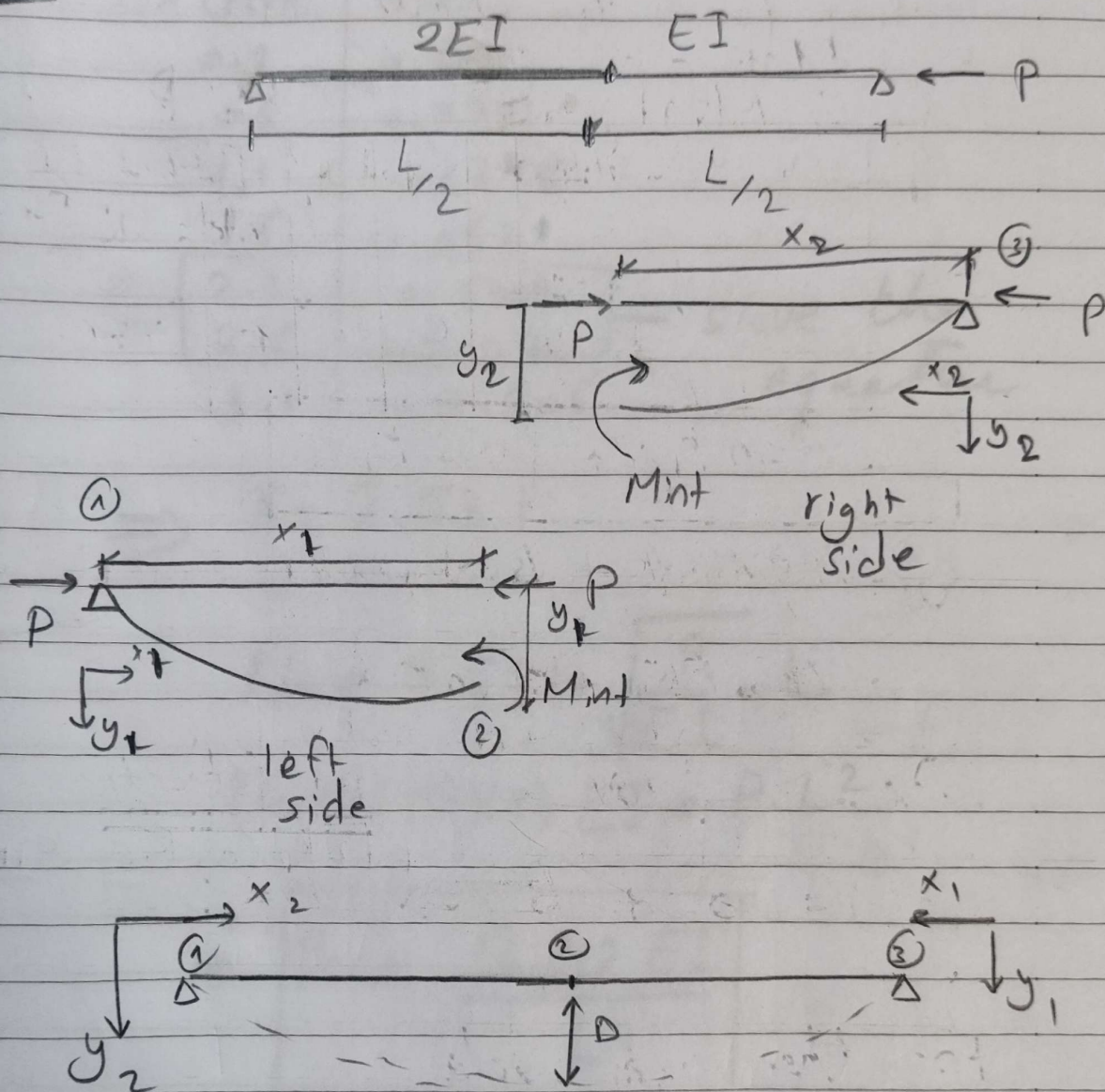


7

التاريخ:

الموضوع:

P2



(8)

التاريخ: / /

(1)

الموضوع:

For the left side

$$Py_1 - M_{int} = 0$$

$$Py_1 - (-2EIy'') = 0$$

$$EIy'' + Py_1 = 0$$

$$k_1^2 = \frac{P}{2EI}$$

$$M_{int} = -2EIy''$$

$$y'' + k_1^2 y = 0$$

$$y_c = A \sin k_1 x_1 + B \cos k_1 x_1$$

$$y_1 = A \sin k_1 x_1 + B \cos k_1 x_1$$

$$y_1' = A k_1 \cos k_1 x_1 - B k_1 \sin k_1 x_1$$

$$\text{at } x_1 = 0 \quad y_1(0) = 0 \quad \underline{\underline{\text{no deflection}}}$$

$$y_1(0) = 0 + B \Rightarrow B = 0$$

For the right side

$$Py_2 - M_{int} = 0$$

$$Py_2 - (-EIy'') = 0$$

$$Py_2 + EIy'' = 0$$

$$EIy'' + Py_2 = 0$$

$$y'' + k_2^2 y_2 = 0$$

$$k_2^2 = \frac{P}{EI}$$

$$M_{int} = -EIy''$$

$$y_c = C \sin k_2 x_2 + D \cos k_2 x_2$$

(9)

التاريخ:

الموضوع:

$$y_2 = C \sin k_2 x_2 + D \cos k_2 x_2$$

$$y_2' = C k_2 \cos k_2 x_2 + D k_2 \sin k_2 x_2$$

$$a) x_2 = 0 \quad y_2(0) = 0$$

$$0 = 0 + D \quad D = 0$$

Remaining 2 constant we need 2 B/c

Use compatibility condition

$$y_1(L/2) = y_2(L/2)$$

$$y_1'(-L/2) = -y_2'(L/2)$$

equation 1

$$A \sin \frac{k_1 L}{2} = C \sin \frac{k_2 L}{2}$$

$$A \sin \frac{k_1 L}{2} - C \sin \frac{k_2 L}{2} = 0$$

equation 2

$$A k_1 \cos \frac{k_1 L}{2} = -C k_2 \cos \frac{k_2 L}{2}$$

$$A k_1 \cos \frac{k_1 L}{2} + C k_2 \cos \frac{k_2 L}{2} = 0$$

(10)

التاريخ: / /

الموضوع:

From the equation (1) and (2) write it in matrix form to have solution:

$$\begin{bmatrix} \sin \frac{k_1 L}{2} & -\sin \frac{k_2 L}{2} \\ k_1 \cos \frac{k_1 L}{2} & k_2 \cos \frac{k_2 L}{2} \end{bmatrix} \begin{bmatrix} A \\ C \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\sin \frac{k_1 L}{2} \times k_2 \cos \frac{k_2 L}{2} + k_1 \cos \frac{k_1 L}{2} \sin \frac{k_2 L}{2}$$

$$\text{from } k_1^2 = \frac{P}{2EI} \quad k_2^2 = \frac{P}{EI}$$

$$k_2 = \sqrt{2} k_1$$

$$\sqrt{2} k_1 \sin \frac{k_1 L}{2} \times \cos \frac{k_1 L}{\sqrt{2}} + k_1 \cos \frac{k_1 L}{2} \sin \frac{k_1 L}{\sqrt{2}} = 0$$

divide everything by

$$\cos \left(\frac{k_1 L}{\sqrt{2}} \right) \cos \left(\frac{k_1 L}{2} \right)$$

$$\sqrt{2} \tan \left(\frac{k_1 L}{2} \right) + \tan \left(\frac{k_1 L}{\sqrt{2}} \right)$$

(11)

التاريخ:

الموضوع:

$x(k, L)$	$f(x)$
0.1	0.141
0.6	0.7741
1.1	1.1248
1.6	1.0621
2.1	0.6009
2.6	-0.102
3.1	-0.806

$$X = 2.531$$

$$K, L = 2.531 = \sqrt{\frac{P}{2EI}} \cdot L$$

$$P_{cr} = \frac{12.812 EI}{L}$$

$$k_e = \sqrt{\frac{P_e}{P_{cr}}} = \sqrt{\frac{\pi^2 EI / L^2}{12.8 EI / L^2}}$$

$$k_e = \sqrt{\frac{\pi^2}{12.8} \cdot 0.88}$$